

Draft Project Report

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Data cleaning and EDA

The Volatility Index, or VIX, has for decades been used as a gauge for market volatility; it is colloquially known as the “fear index,” because it is strongly connected to fear in the financial markets. It measures the 30-day implied volatility of SPX options, where SPX is the S&P 500 index.

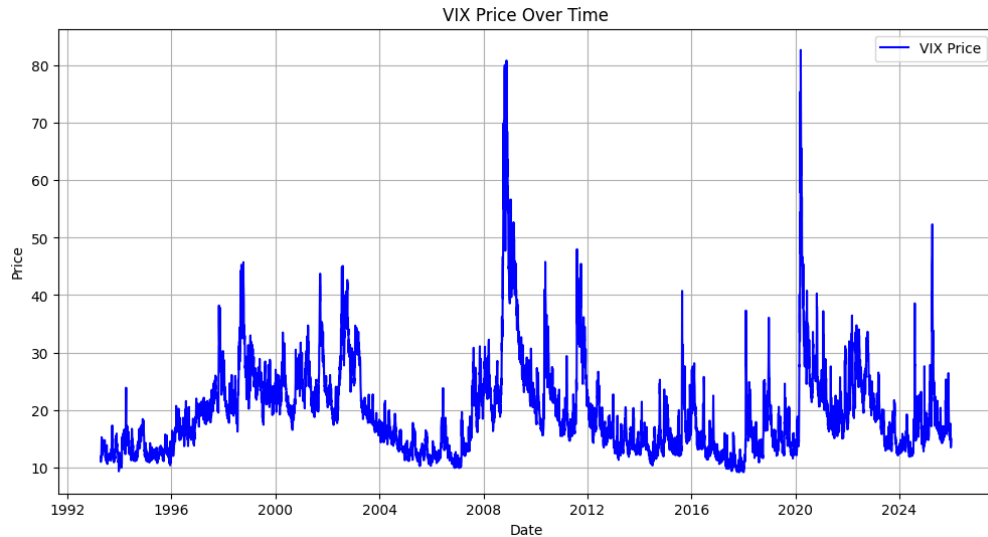


Figure 1: VIX price since 1993

The VIX is inversely correlated with the broader market returns ($r = -.71$, $p < .001$ between 1993 and 2026 for monthly returns): when the market goes up, the VIX tends to fall. Conversely, when the market sharply downturns, the VIX tends to shoot up, hence the term “VIX spike.” Looking at the plot above, you see VIX spikes during times of macroeconomic instability: The 2008 Financial Crisis, COVID in 2020, tariff uncertainty in 2025, among other things. Looking at the plot below illustrates the contrast between the VIX and the S&P 500, visually indicating that the monthly S&P 500 returns tend to oppose monthly VIX returns.

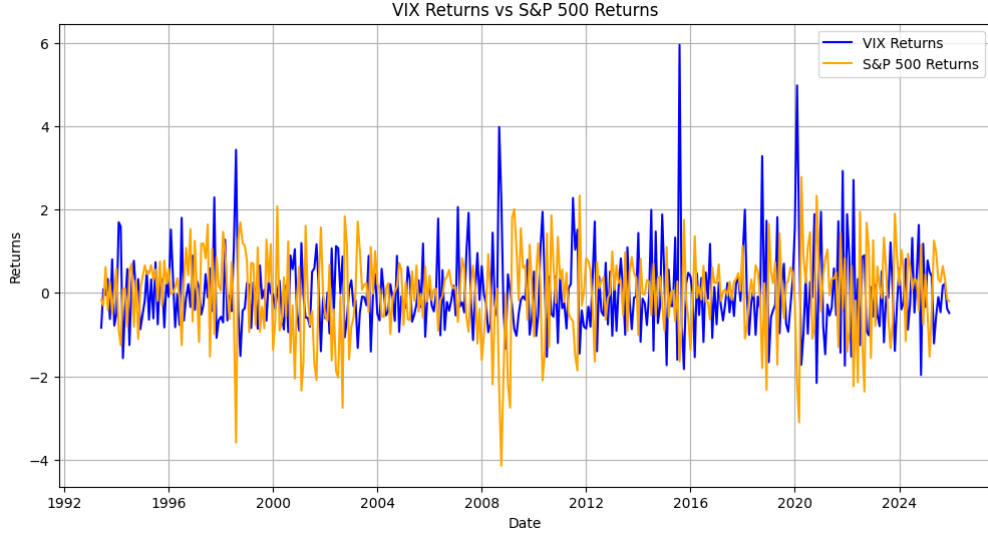


Figure 2: Standardized VIX and S&P500 returns since 1993

We will use the following data for our analysis: 1. Reddit comments from r/wallstreetbets, analyzed for sentiment using roBERTa, an NLP model. 2. S&P 500 daily returns. 3. Geopolitical Risk Index data, which tracks the frequency of adverse geopolitical events as reported in 10 large newspapers. Using this data, **we will predict daily VIX log returns.**

For the Reddit sentiment feature, raw r/wallstreetbets comments were scored with roBERTa and then aggregated to a daily table. The resulting sentiment file contains one row per `created_date`, with `final_score` representing the daily sentiment signal and `score` representing the average Reddit comment score for that date. We convert `created_date` to a date field, merge the daily sentiment data with market and GPR observations by date, drop dates where the merged modeling inputs are unavailable, and lag `final_score` by one trading day before using it in the RNN so the model does not use same-day sentiment to predict the same day's VIX close.

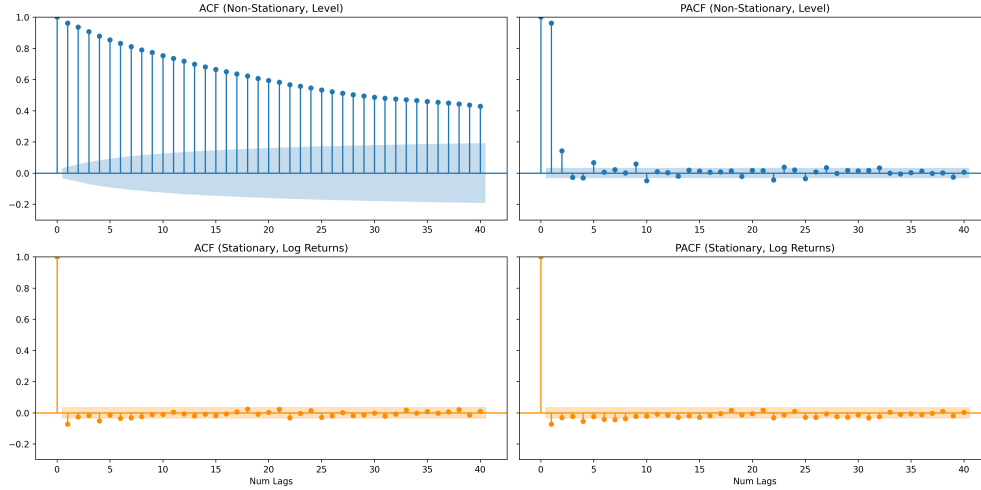


Figure 3: ACF and PACF for VIX (Level) and VIX (Log Transformed)

Figure 3 shows the autocorrelation (left) and partial autocorrelation (right) of the VIX and VIX log returns. The figure indicates that the VIX is dependent on the days that come before it, especially the previous day. However, the log returns show much weaker evidence of autocorrelation.

The cleaned modeling table is built by aligning observations by date, sorting chronologically, and dropping rows where merged sources or engineered lags are unavailable. The target is close-to-close VIX log return, and all exogenous variables used for prediction are shifted or lagged so the models only use information available before the close being predicted.

SARIMAX data load and merge

The SARIMAX workflow starts with two daily data sources:

1. [Geopolitical Risk Index](#)
2. Volatility Index (VIX)

	date	GPRD	GPRD_ACT	GPRD_THREAT
0	1985-01-01	230.039429	275.197601	153.027985
1	1985-01-02	115.676971	146.772064	87.444572
2	1985-01-03	97.428459	158.937653	29.459934
3	1985-01-04	157.366638	156.882248	157.026352
4	1985-01-05	81.361313	92.698143	77.319405

	date	adjclose
0	2012-01-03	22.969999
1	2012-01-04	22.219999
2	2012-01-05	21.480000
3	2012-01-06	20.629999
4	2012-01-09	21.070000

	date	adjclose	GPRD	GPRD_ACT	GPRD_THREAT	log_return
1	2012-01-04	22.219999	76.942284	48.445545	94.286194	-0.033196
2	2012-01-05	21.480000	82.439468	44.828571	112.174370	-0.033871
3	2012-01-06	20.629999	100.598587	50.144466	137.426437	-0.040376
4	2012-01-09	21.070000	88.391258	58.746124	105.000153	0.021104
5	2012-01-10	20.690001	66.643372	31.265053	84.753967	-0.018200

Feature engineering and selection

Stationarity

If we want to forecast a time series using ARMA-type models, it needs to be stationary first.

A time series is stationary when it has:

1. a constant mean.
2. a constant variance.
3. a constant autocorrelation.

If a time series is not stationary, we can apply differencing (or other transforms) to remove trends or seasonality.

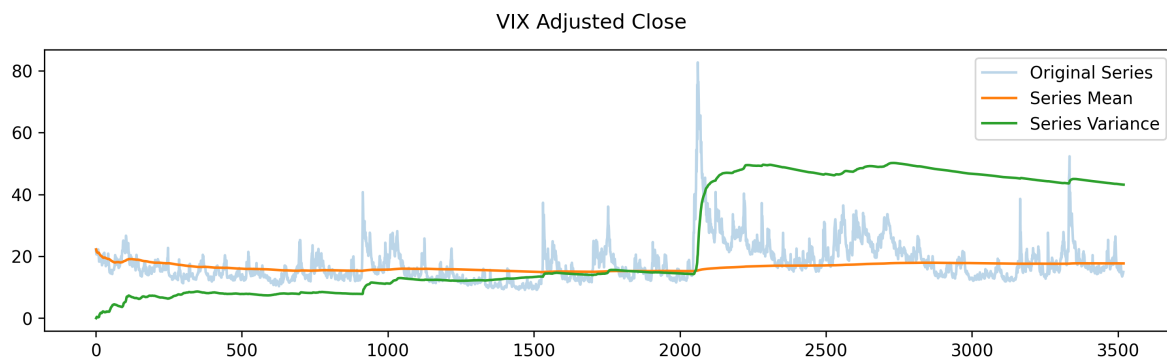
$$\Delta Y_t = Y_t - Y_{t-1}$$

First we will perform the **Augmented Dickey-Fuller Test**.

Rejecting the null hypothesis ($P \leq 0.05$) essentially is saying, “This series has a level that it always tends to drift towards.” - it is stationary.

p = 5.552993E-13, ts = -8.2460

The VIX's adjusted close is likely stationary according to `adfuller`. Let's visualize it anyway:



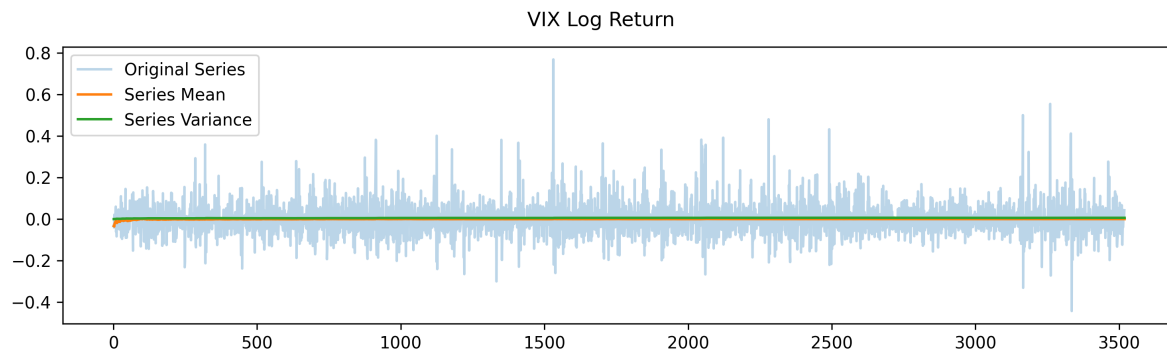
The series mean is mostly flat, albeit shaky. The series variance has a massive regime shift around $t = 2050$; which likely violates stationarity.

Now we can try diffing (log return) to see if that yields any improvement.

***Aside:** Another option would be splitting the data into two different regimes before and after the shift and training a model on either regime.*

`p = 0.000000E+00, ts = -63.7583`

The test statistic is almost 8 times greater in magnitude: a major improvement. Let's visualize:



Wow! Almost perfectly flat. This series is most certainly stationary. We are now able to be confident that ARMA-type models can reliably forecast the *direction* of the series, but to ensure that magnitude forecasts are accurate, we will need the help of another test.

Engle's Test for Autoregressive Conditional Heteroscedasticity tells us if variance regimes are sticky.

Rejecting the null hypothesis tells us that volatility tends affect future prices and ARMA-type models will need some help forecasting swing magnitudes.

p = 2.077737E-32, ts = 175.4318

Predictor viability

Can we actually use geopolitical risk to predict the VIX's log returns? The **Granger Causality Test** can tell us just that.

Rejecting the null hypothesis tells us that the exogenous series at a given lag is a valid predictor of the VIX's log returns.

Granger Causality

number of lags (no zero) 1

ssr based F test:	F=5.6852	, p=0.0172	, df_denom=3515, df_num=1
ssr based chi2 test:	chi2=5.6901	, p=0.0171	, df=1
likelihood ratio test:	chi2=5.6855	, p=0.0171	, df=1
parameter F test:	F=5.6852	, p=0.0172	, df_denom=3515, df_num=1

Granger Causality

number of lags (no zero) 2

ssr based F test:	F=3.4773	, p=0.0310	, df_denom=3512, df_num=2
ssr based chi2 test:	chi2=6.9645	, p=0.0307	, df=2
likelihood ratio test:	chi2=6.9576	, p=0.0308	, df=2
parameter F test:	F=3.4773	, p=0.0310	, df_denom=3512, df_num=2

Granger Causality

number of lags (no zero) 3

ssr based F test:	F=2.9483	, p=0.0316	, df_denom=3509, df_num=3
ssr based chi2 test:	chi2=8.8626	, p=0.0312	, df=3
likelihood ratio test:	chi2=8.8515	, p=0.0313	, df=3
parameter F test:	F=2.9483	, p=0.0316	, df_denom=3509, df_num=3

Granger Causality

number of lags (no zero) 1

ssr based F test:	F=3.2129	, p=0.0731	, df_denom=3515, df_num=1
ssr based chi2 test:	chi2=3.2156	, p=0.0729	, df=1
likelihood ratio test:	chi2=3.2142	, p=0.0730	, df=1
parameter F test:	F=3.2129	, p=0.0731	, df_denom=3515, df_num=1

Granger Causality

number of lags (no zero) 2

ssr based F test: F=2.6284 , p=0.0723 , df_denom=3512, df_num=2
 ssr based chi2 test: chi2=5.2643 , p=0.0719 , df=2
 likelihood ratio test: chi2=5.2604 , p=0.0721 , df=2
 parameter F test: F=2.6284 , p=0.0723 , df_denom=3512, df_num=2

Granger Causality

number of lags (no zero) 3

ssr based F test: F=2.2327 , p=0.0824 , df_denom=3509, df_num=3
 ssr based chi2 test: chi2=6.7114 , p=0.0817 , df=3
 likelihood ratio test: chi2=6.7050 , p=0.0819 , df=3
 parameter F test: F=2.2327 , p=0.0824 , df_denom=3509, df_num=3

Granger Causality

number of lags (no zero) 1

ssr based F test: F=5.8156 , p=0.0159 , df_denom=3515, df_num=1
 ssr based chi2 test: chi2=5.8206 , p=0.0158 , df=1
 likelihood ratio test: chi2=5.8158 , p=0.0159 , df=1
 parameter F test: F=5.8156 , p=0.0159 , df_denom=3515, df_num=1

Granger Causality

number of lags (no zero) 2

ssr based F test: F=2.9694 , p=0.0515 , df_denom=3512, df_num=2
 ssr based chi2 test: chi2=5.9472 , p=0.0511 , df=2
 likelihood ratio test: chi2=5.9422 , p=0.0512 , df=2
 parameter F test: F=2.9694 , p=0.0515 , df_denom=3512, df_num=2

Granger Causality

number of lags (no zero) 3

ssr based F test: F=2.3776 , p=0.0680 , df_denom=3509, df_num=3
 ssr based chi2 test: chi2=7.1471 , p=0.0674 , df=3
 likelihood ratio test: chi2=7.1399 , p=0.0676 , df=3
 parameter F test: F=2.3776 , p=0.0680 , df_denom=3509, df_num=3

Lagged and rolling features

The following functions create lagged features (shifted according to the granger results) for the exogenous columns (geopolitical risk) and for log return (rolling statistics).

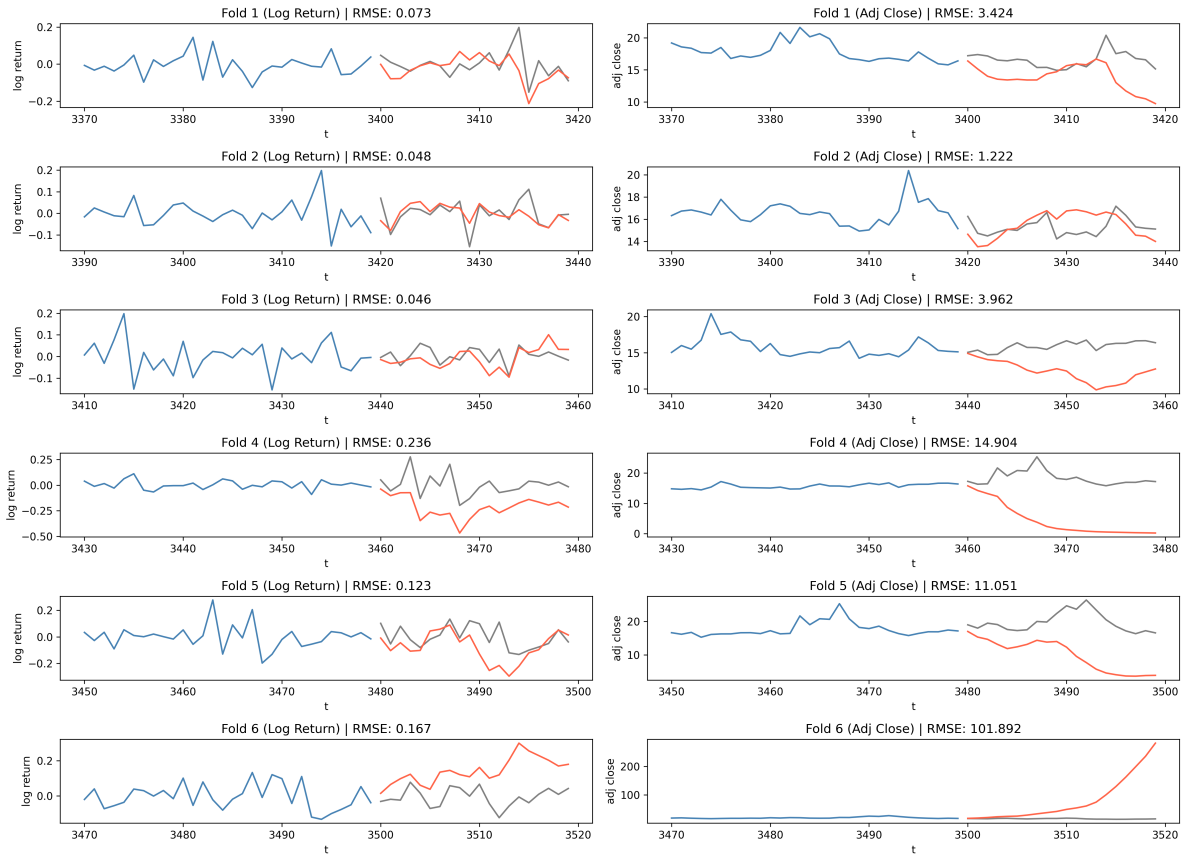
Multiple models

SARIMAX model

The SARIMAX model provides a classical time-series benchmark. It uses lagged geopolitical risk variables as exogenous regressors and lagged rolling VIX log-return statistics as autoregressive features.

Walk-forward SARIMAX

SARIMAX (7, 1, 2) | RMSE (Return): 0.116 | RMSE (Close): \$22.742



As expected, the model is quite **accurate at forecasting timing and direction of swings**, but can be **very wrong in terms of magnitude**.

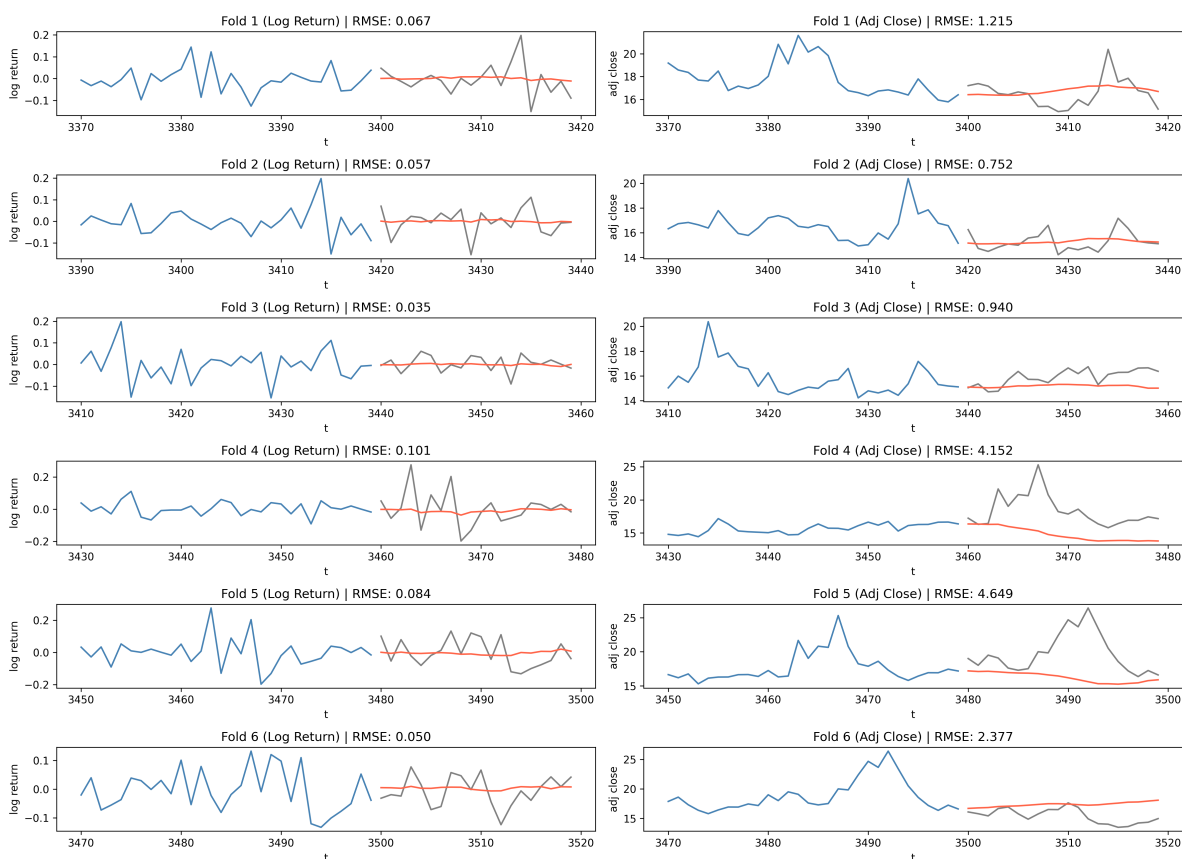
This is especially noticeable in folds 4 and 6, where the log return forecast looks mostly accurate, but upon transforming back to closing price, the predictions completely break down.

Hyperparameter tuning

0% | 0/5 [00:00<?, ?it/s]

```
[I 2026-05-27 23:22:57,841] Trial 0 finished with value: 8.125436249318147 and parameters: {}
[I 2026-05-27 23:23:02,395] Trial 1 finished with value: 15.206125713737956 and parameters: {}
[I 2026-05-27 23:23:13,247] Trial 2 finished with value: 6.728785153337213 and parameters: {}
[I 2026-05-27 23:23:16,699] Trial 3 finished with value: 2.338328926459994 and parameters: {}
[I 2026-05-27 23:23:26,429] Trial 4 finished with value: 265380186.13997746 and parameters: {}
Best value (mean RMSE price): 2.338328926459994
Best params: {'p': 1, 'd': 0, 'q': 1, 'win_3': True, 'win_5': True, 'win_10': False, 'win_15': False}
Best (p,d,q): (1, 0, 1)
Best features: [(3, '3d'), (5, '5d'), (30, '30d')]
```

SARIMAX (7, 1, 2) | RMSE (Return): 0.066 | RMSE (Close): \$2.348



Even though the metrics are much better in this “finetuned” model, upon plotting, we see that our model loses its ability to predict swings. Instead, the model seems to predict the mean of

the series. While it is accurate in this sense, perhaps RMSE is not the best metric if we want to predict swings.

RNN model

The RNN notebook (`vix/analyses/rnn.ipynb`) extends the SARIMAX workflow by adding Reddit sentiment and S&P 500 inputs to the VIX and geopolitical-risk data. The merged table contains daily VIX adjusted close, S&P 500 adjusted close, r/wallstreetbets sentiment (`final_score`), and the daily GPR indexes. VIX and S&P 500 close-to-close log returns are created after the merge.

The RNN feature set is built to avoid leakage. GPR variables and Reddit sentiment are shifted by one trading day, and the market inputs are represented with explicit 1-, 2-, 5-, and 10-day lags for both VIX and S&P 500 log returns and adjusted closes. The notebook also adds 21-day, 42-day, and 365-day rolling means and standard deviations for shifted VIX log returns and shifted S&P 500 log returns.

The model predicts one day at a time from a 20-day lookback window. Each fold fits `StandardScaler` objects on training rows only, reserves the last 200 training targets for validation, and evaluates with `TimeSeriesSplit(n_splits=10, test_size=200)`. The network is a Keras GRU with 64 hidden units, layer normalization, a GELU dense layer, and a single linear output. It is trained with AdamW when available, mean-squared-error loss, up to 75 epochs, and early stopping with patience 10.

The RNN's mean log-return RMSE is 0.0872, with mean MAE 0.0653 and mean directional accuracy 0.5015. This is directionally close to a coin flip and worse than the zero-return and train-mean baselines on RMSE within the RNN notebook's 10-fold setup. When converted back to VIX prices, the one-day anchored price RMSE is 2.1703, while the recursive price path is unstable because small log-return errors compound over the full prediction path.

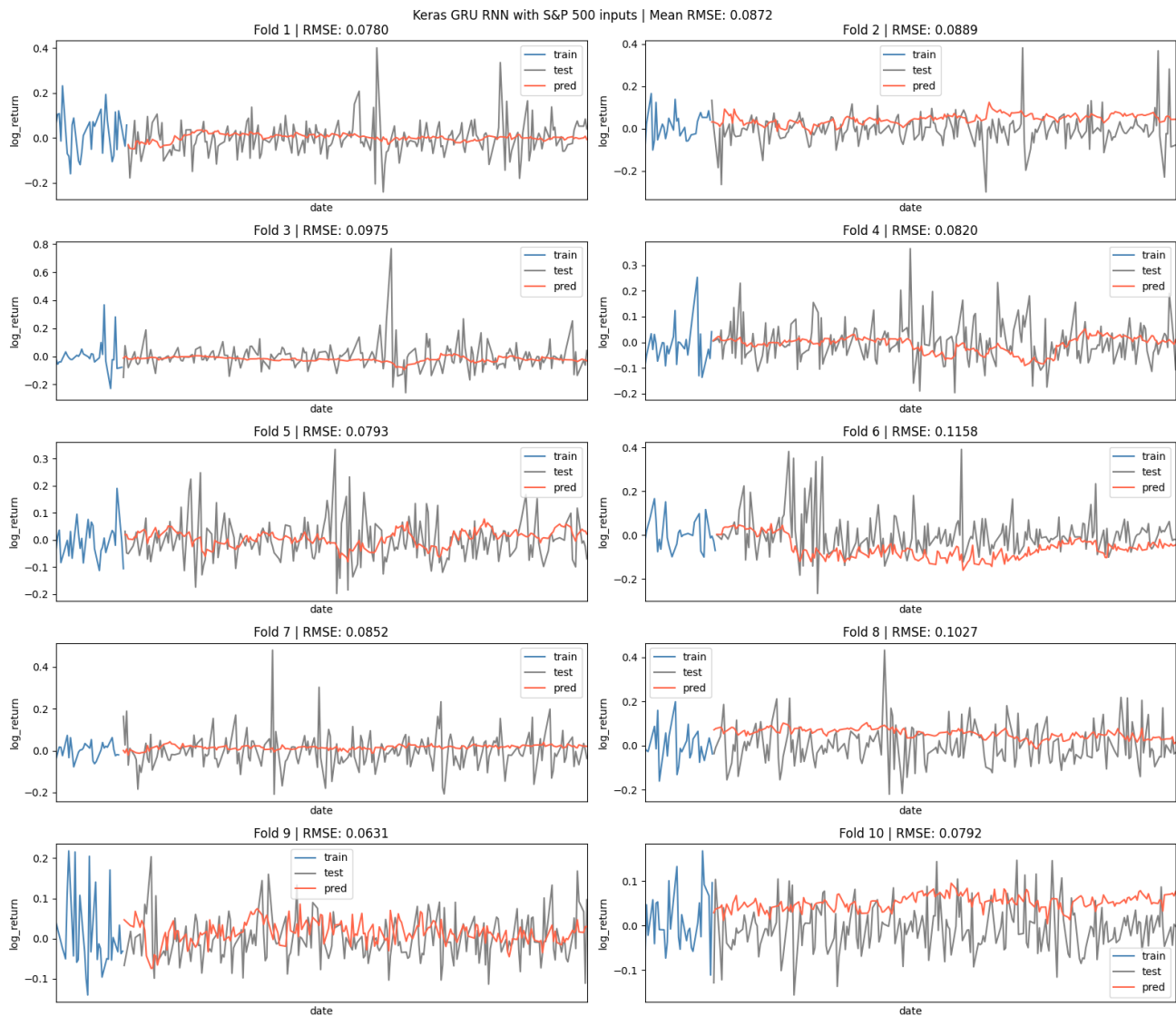


Figure 4: Keras GRU fold-level walk-forward predictions

Figure 4 shows the fold-level walk-forward forecasts from the RNN notebook. The model generally produces smoother forecasts than the realized VIX log returns, which helps explain why it struggles with spike magnitude even when it sometimes tracks the broad direction.

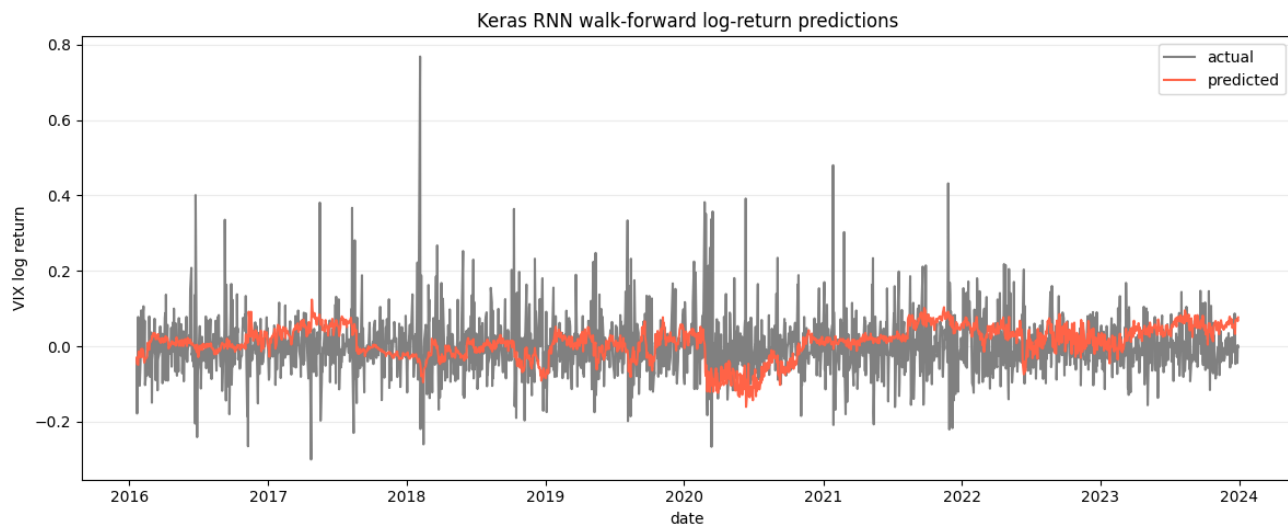


Figure 5: Keras GRU actual versus predicted VIX log returns

Figure 5 pools the RNN predictions across the walk-forward folds. The predicted series is visibly less volatile than the actual VIX log-return series, reinforcing the numeric result that the model does not yet outperform simple centered-return baselines.

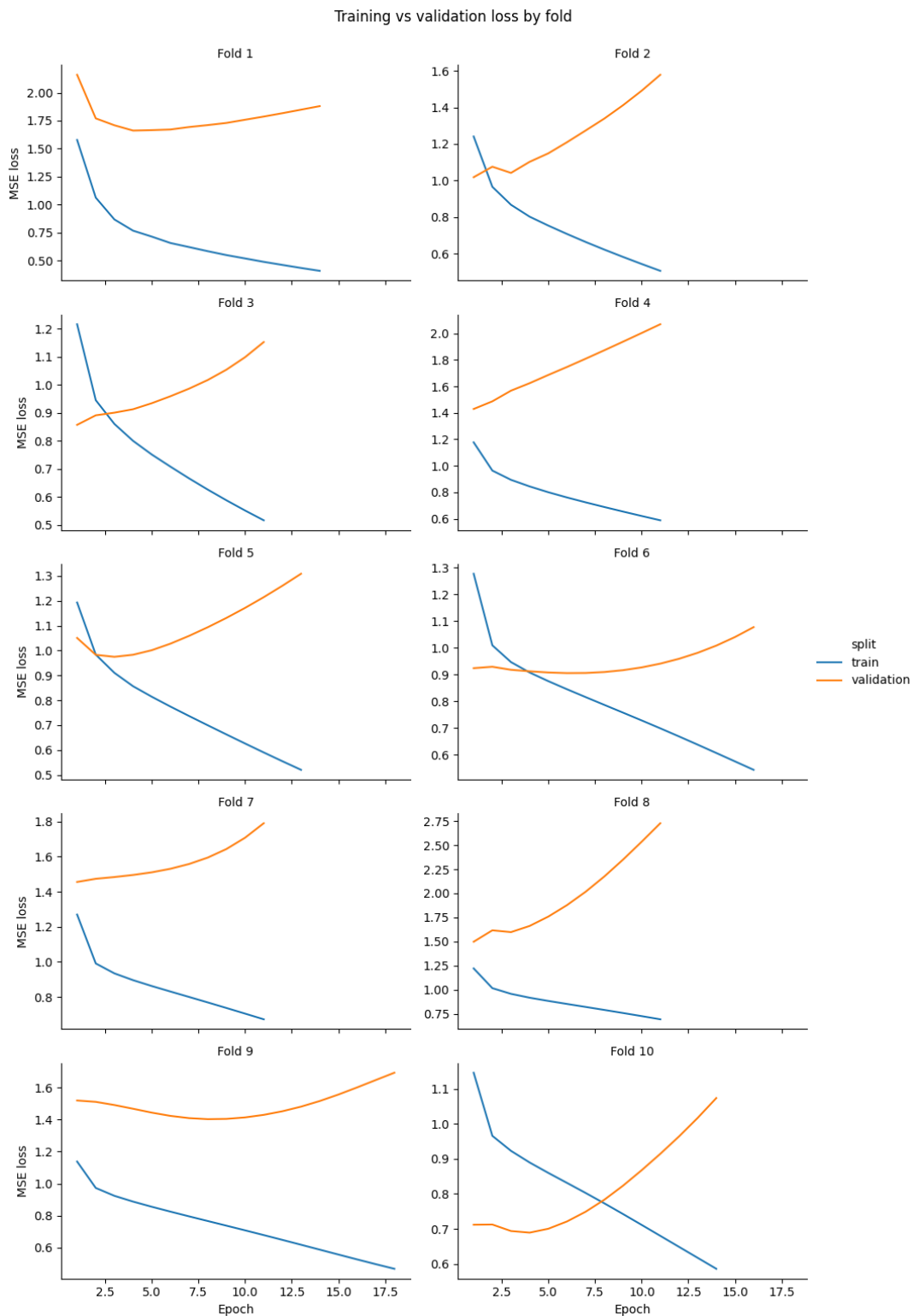


Figure 6: Keras GRU training and validation loss by fold

Figure 6 shows the GRU’s training and validation losses across folds. The validation loss often flattens or rises quickly while training loss continues improving, which suggests the current RNN specification is prone to overfitting rather than learning a stable out-of-sample volatility signal.

Initial results

The custom SARIMAX model is able to forecast some swing timing and direction, but it can be very wrong in terms of magnitude. This is especially noticeable in folds where the log-return forecast looks reasonable, but the transformation back to closing price compounds the errors.

The finetuned SARIMAX model improves the headline RMSE, but its predictions become much more mean-reverting. This makes it look better under RMSE while reducing its usefulness for detecting volatility spikes.

The GRU RNN adds a richer set of lagged market and sentiment features, but its initial results do not beat the simple zero-return or train-mean baselines in the notebook’s evaluation. Its main benefit at this stage is as a flexible modeling framework that can incorporate longer sequential context, but the current specification does not yet translate that flexibility into better error metrics.

Preliminary comparisons

Model	Mean RMSE (log returns)	Mean MAE (log returns)	Directional accuracy	Notes
SARIMAX (custom)	0.116	–	–	Better at visible swing timing than magnitude.
SARIMAX (finetuned)	0.066	–	–	Lower RMSE, but tends to predict the mean.
Keras GRU RNN	0.087	0.065	0.502	Uses Reddit sentiment, GPR, S&P 500, and VIX lag/rolling features.

Model	Mean RMSE (log returns)	Mean MAE (log returns)	Directional accuracy	Notes
Zero-return baseline (RNN folds)	0.077	0.055	0.004	Strong RMSE baseline because daily VIX log returns are centered near zero.
Previous VIX return baseline (RNN folds)	0.114	0.081	0.487	Worse than the GRU on RMSE and direction.

These comparisons are preliminary because the SARIMAX and RNN evaluations currently use different cross-validation windows: the SARIMAX workflow uses 6 splits with 20-day test windows, while the RNN notebook uses 10 splits with 200-day test windows.

Early conclusions

Overall, time series data is a great challenge to model properly. From designing autoregressive features to ensuring no data leakage, forecasting requires a high degree of attention-to-detail and awareness of the tools available.

In terms of modeling, we find that more complex models that might work better for tabular data are not immediate winners in forecasting. In fact, classic statistical methods tend to dominate this first pass. The next step is to standardize the evaluation windows across models and consider metrics beyond RMSE, especially if the project goal is to identify VIX spikes rather than only minimize average daily forecast error.